Methods of Applying Flues in Locomotive Boilers*

This report, which was read at the annual meeting of the Master Boiler Makers' Association held at Detroit, May 23 to 26, was compiled from the flue practice of the more important railroads of the country. It includes details of how flue ends should be prepared, the manner of preparing holes in the tube sheets, the type and size of copper ferrules to be used and the method of tightening flues in the sheets, as well as the standard tools for rolling, prossering and beading flues.

AFTER checking up the methods of various railroads, it is believed that for a two inch diameter flue, flues should be swedged 1¾ inches diameter for a distance of seven-eighths of an inch straight, then tapered back to nothing for a distance of 3⅛ inches. This is far superior to a flue swedged 1⅞ inches diameter straight for one inch, with a taper of one-quarter to its original diameter, to provide increased water space, for it is a well recognized fact that the first few inches of the flue measured from the flue sheet, is the most efficient heating surface. This increased water space allows the steam bubbles to form and pass up between flues unrestricted. If it is not provided, the bubbles are

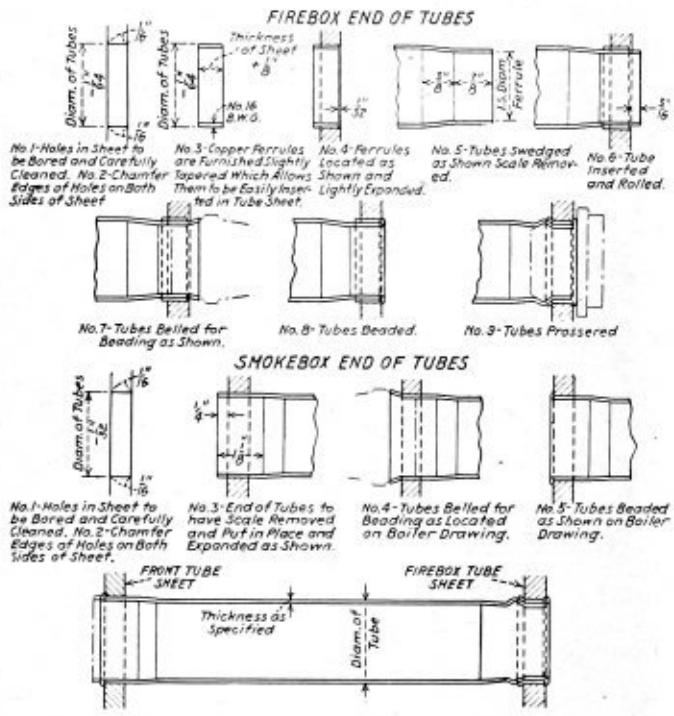


Fig. 1

trapped between the flues and drive the water away from the flue sheet. This causes the flue sheet and flue ends to overheat. This is one of the best known methods of shortening the life of flues. After being swedged flues should be annealed and care taken to see that all scale is removed from that part of the flue which is to enter the sheet, for it is a well known fact that we are unable to get tight flues with scale or any dirty substance between the copper ferrule and flue.

The swedging of superheater flues seems to be the general practice on all our railroads, the swedge being of the following dimensions: In 5½-inch outside diameter flues, 4½ inches outside diameter, 5⅛ inches straight, tapering to the 5½-inch diameter, three inches back; flues to be prepared as to the removal of scale the same as on smaller flues.

*Report prepared by a committee consisting of A. F. Stiglmeier, chairman; E. C. Umlauf and James F. Walsh.

Preparation of Flues for Installation

The preparing of flues, both small and superheater for the front tube sheet, seems to lie between the practice of shimming the same with tin, or expanding to fit the hole. Expanding should be done before the flue leaves the flue department, for good results can be obtained when heated properly, for there are many simple ways that are not harmful to flues. In shimming flue with tin, there is a possible chance that when the boiler maker applies the shim he will not get it through the sheet, so that it may blow out, causing a failure besides possible damage.

Drilling flue holes in the back flue sheet will have to be governed by the diameter of the swedge and copper ferrule used. The committee recommends that all flue holes in back flue sheet have inside and outside edges slightly rounded, to remove the sharp edges, the radius to be about 1/16 of an inch. Flue holes in old back flue sheet should be examined for sharp edges, rust and scale and the same removed. Holes 1/32 inch or more out of round, should be reamed. In this you will readily see the advantages of a 1¾-inch diameter swedge over a larger size. The smaller flue hole will give increased bridge strength between the flues, and at this time, when it seems to be the general practice on most of the railroads to electric weld the beads of flues to the back flue sheet, a greater space is required between the flue beads. If not provided, the electric welds will over-lap each other, which is not a good practice.

Drilling holes in the front flue sheet seems to be a general practice as to the diameter on all our railroads. The committee recommends that all holes drilled for flues smaller than superheater be 1/16 of an inch larger than the outside diameter of flues, and for superheater flues three thirty-seconds larger than the outside diameter of flues. Holes

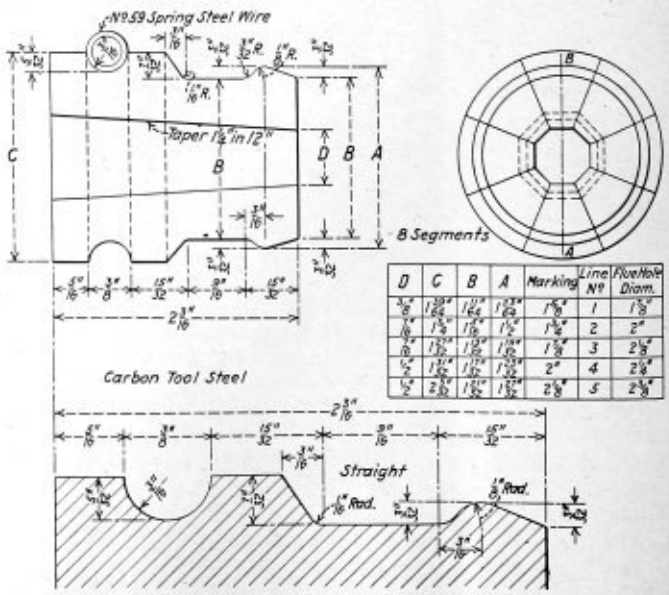


Fig. 2

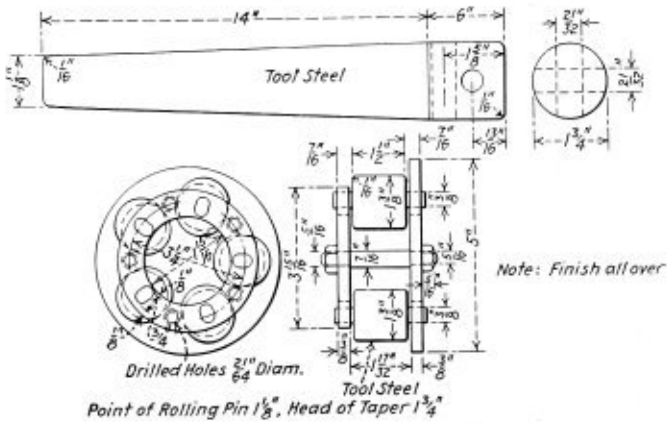


Fig. 7

back flue sheet, sectional expanders be used to expand them inside of the sheet. They are then to be rolled in the front end. Flues should not be beaded at the front or back end until after the boiler is tested. Investigation of different cases of leaking flues has shown that by the other method of testing leaking flues may be covered up. If a flue tested before being beaded shows no leak it will not developed a leak after being in service a short while.

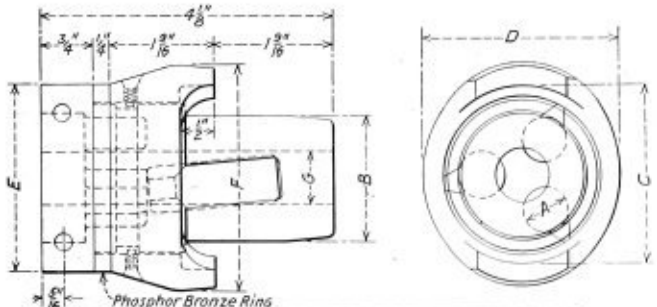
Figs. 1 to 8 inclusive show sketches of standard tools for rolling, prossering and beading tools, as well as various operations incident to flue installation.

The committee also recommends that a flue roller should never, under any circumstances, be used in a flue after it has been expanded with a sectional expander. It will stretch the metal and break the joint made by the sectional expander and beading tool.

Applying Flues in Stationary Boilers

As to the application of flues to stationary boilers, the chairman corresponded with various boiler manufacturers and among replies received were the following:

"In applying flues to water leg type of watertube boiler, using four inch tube expanded into one-half inch plate in each boiler leg, we first roll the tube in position with three parallel rolls; then take out one of the rolls and put in a roll designed to bell the end of the tube which extends into the water leg. The tube end is thus belled or flared to comply with the A.S.M.E. Boiler Code. We are now experimenting with another type of expander body and have had



G	F	E	D	C	B	A	SIZE	USED ON	LINE NO.
21/32"	2 5/8"	2 1/16"	2 3/8"	2 3/8"	1 3/8"	1/2"	1 5/8"	2" Tubes Rear New Work	1
21/32"	2 5/8"	2 5/16"	2 1/2"	2 1/4"	1 1/2"	3/8"	1 3/4"	2" Tubes Rear Repair	2
23/32"	3"	2 5/16"	2 5/8"	2 3/8"	1 59/64"	19/32"	1 7/8"	2" Tubes Front New Work 2 1/8" Tubes Rear New Work	3
25/32"	3 1/8"	2 9/16"	2 3/4"	2 1/2"	1 63/32"	5/8"	2"	2" Tubes Front Repair 2 1/8" Tubes Rear Repair	4
25/32"	3 1/4"	2 5/16"	2 7/8"	2 5/8"	1 15/16"	11/16"	2 1/8"	2 1/8" Tubes Front New Work	5
25/32"	3 3/8"	2 1/16"	3"	2 3/4"	1 15/16"	3/4"	2 1/4"	2 1/4" Tubes Front Repairs	6

Fig. 8

good results, so far. This body has three rolls, two of which are slightly tapered so that the surface of the tube as rolled will be parallel to the axis of the tube—in other words, the outside edge of the rolls, where it comes in contact with the tube, is exactly at right angles to the tube sheet. The other roll in this new type of expander is designed to bell or flare the end of the tubes, thus the rolling and flaring is done at the same time."

Tools for Expanding and Working Flues in Stationary Boilers

"In answer to your inquiry we are attaching herewith two prints showing general type of tools used in expanding the ends of tubes in Heine boilers. The sheet which contains a picture of a Heine boiler also shows the general type of tool which is used by power plants for the renewal of individual tubes when it becomes necessary. The other print shows the general type of expander used in the shop. You will note that this latter tool is more elaborate in that it has a collar which acts as a guide for the mandrel as well as to take the thrust caused by the rollers when the mandrel is driven in place. In Heine boilers it is not required that the ends of the tubes be beaded, but it is necessary to flare them. The American Society of Mechanical Engineers Boiler Code,

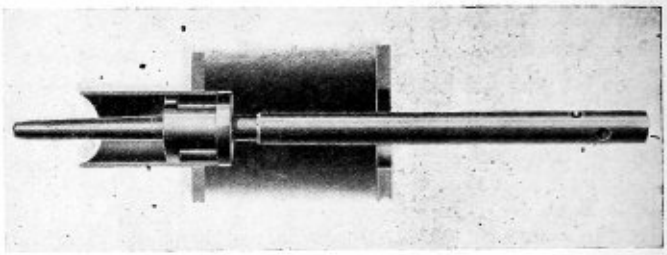


Fig. 9

under which we construct most of our boilers, calls for the tube ends to be flared to at least one-eighth inch over the diameter of the tube hole. Ordinary roller expanders have a tendency to slightly flare the projecting ends when the rolling is done, but a slight additional flare is provided by driving a conical piece into the tube end holes, spreading it the required amount. One or two licks of a comparatively heavy hammer will do this flaring without loosening the tube where it has been rolled."

Tools for stationary boiler tubes are shown in Figs. 9 and 10.

"The great majority of boilers we manufacture are of the fire tube type for stationary purposes. It has always been our custom to use a prosser expander for putting tubes in fire tube boilers; after the tubes have been expanded by means of the prosser we bead the tube against the sheet. We allow 1/64 inch clearance between tube and sheet. Some times we are requested to provide the rear tube sheet with copper ferrules. Where copper ferrules are used we follow the same procedure and use the prosser expander. Some manufacturers simply roll the tubes in the sheet, but we do not consider it good practice as we feel that in a fire tube type of boiler the tube is acting more or less as a stay and by expanding the tube by means of a prosser expander that portion of the tube expanded inside the tube plate acts more or less as a lock nut would act on a stay rod. In building boilers of the watertube type we, of course, roll the tube and after rolling flare the end of the tube."

"1. Tubes for stationary watertube, and flues for fire tube steam boilers shall conform as to specifications of manufacture and thickness to the following instructions and shall be requisitioned in accordance therewith.

that this association, since we have had so many broken bolts in combustion chamber engines, those equipped with flexible bolts, particularly, that we go on record that we hammer test flexible staybolts each time we test the other staybolts.

W. J. Murphy, Pennsylvania System: Mr. President, I think that is the rule now on all railroads, that you test all staybolts when you go into the firebox, hitting all the bolts in the firebox.

H. J. Wandberg, Chicago, Milwaukee and St. Paul: On our railroad, we hammer test all staybolts, whether flexible or not, but we have the end of our flexible staybolts marked with red so that we can distinguish them from the rigid staybolts. I think most railroads are carrying out that practice of testing all staybolts.

A. W. Schelton: On the Frisco System we have no flexible staybolts marked and every bolt is hammer tested once a month—all of them. We are of the opinion that we find almost as many flexible staybolts broken as we do rigid staybolts in our monthly tests. We find those broken bolts by going over them with at least one hundred pounds of steam, and then we cool the engine and go over the rest of the bolts.

Charles P. Patrick, Meadville Machinery Company: I want to make an explanation of my motion. Some railroads make tests on flexible bolts, but the law requires you to test those bolts only once a year under pressure, and so all are not tested. Now, in the company with which I am associated, we test our bolts every month. If an inspector knows that a combustion chamber is equipped with flexible staybolts and that he is not required to test them, he is liable not to do so each month, since the law requires taking off the caps every twenty-four months. We find broken flexible bolts under the hammer test. We find them broken every month. It is only a short time ago that we found four in one engine and four in another under the hammer test. Even if you do not do so usually, if you suspect the bolt is broken, from the hammer test, it will pay you to remove the cap.

L. E. Hart, Atlantic Coast Lines: We have a class of engines equipped with flexible staybolts in the combustion chamber, and we hammer test them and find lots of them broken. We hammer test all of the bolts, and whenever we have the expansion stays in front, we find some of them broken.

A. N. Lucas, Oxweld Railroad Service: Is it not a fact that the federal boiler law and rules requires us to test all bolts every month? According to the rule, the cap shall be removed every twenty-four months, but does it not say we should test the bolts every month?

E. W. Young, Chicago, Milwaukee and St. Paul: Rule Number Sixteen states: "Method of testing flexible staybolts having caps. All flexible staybolts having caps over the outer ends shall have the caps removed at least once every two years and also whenever the United States inspector or the railroad company's inspector considers the removal desirable in order to thoroughly inspect the staybolts. The firebox sheets should be examined carefully at least once a month to detect any bulging or indications of broken staybolts. Each time a hydrostatic test is applied, the hammer test required by rules 21 and 22 shall be made while the boiler is under hydrostatic pressure not less than the allowed working pressure and proper notation of such test made on Form Number Three."

In order to overcome any mistakes we instruct our men, where it says, "Firebox sheets should be examined carefully at least once a month to detect any bulging or indications of broken staybolts," to examine flexible as well as rigid staybolts, and they cannot examine sheets without testing each and every staybolt. They will detect bulging or any other flaws when they examine the sheets carefully, and we instruct them to test all staybolts.

Charles P. Patrick: It is not in the rule to hammer test staybolts.

E. W. Young: No, but to overcome the trouble in the sheet, we must examine the bolts, and there you have the rule.



Thomas Lewis, Retiring President



E. W. Young, President



Frank Gray, First Vice-President

W. J. Murphy, Pennsylvania System: I have had some little experience in testing staybolts and here is what we have found in testing flexible staybolts. We have a medium sized hammer for testing the ordinary staybolts and about twice the size for testing flexible bolts. We have found that by using a larger hammer we get better results. The question came up as to whether you find flexible bolts broken or not, and I will tell you that you can find them by hammer testing them. Any good inspector will find broken flexible staybolts, but he will get better results with a larger hammer.

R. C. Young, Chicago and North Western: I would like to know if a flexible staybolt, when the boiler is not under pressure, is not in the same condition as other broken staybolts? It is loose, not bent nor separated, and there is not much success from testing flexible staybolts unless you have pressure on the boiler and tension on the bolts, because when there is no pressure in the boiler, it seems to me much the same as when a bolt is broken.

T. W. Lowe, Canadian Pacific: We find in the law, as agreed, that there is nothing in it which tells us that we cannot do something better than what the law suggests. Therefore, on the western line of the Canadian Pacific, where we use the flexible staybolts, in general, we find it very advisable to have water pressure to the extent of 140 pounds at every test, every thirty days' test—shop test or

when applying the flues. Seventy-five percent of the railroads, from whom we have answers, are applying them with the sectional expander, getting away from the roller expander, and I believe that is what is cracking our flues.

F. L. Biklen, C. B. & Q.: I want to say that the C. B. & Q. has done away with the rolling in the back end since 1904. With copper ferrules we use a standard, now, that is one inch wide and that has been our practice since 1904, and we have found it very satisfactory.

W. Wallough, Omaha: I have used the sectional expander for the last few years and never split the steel flues that I remember. We have split iron flues. You cannot back them out of the sheet and you do not need to roll them when you get through.

C. F. Young, Chicago and North Western: There is one feature about this, where the holes are oblong, after the holes are reamed out, the diameter of the holes increases to the same size as the ferrule. That is, practically, reducing the gage of the flue. I think if the holes are larger in diameter, that the thickness of the ferrules should be increased to maintain the flue to its original gage.

D. A. Lucas: We use copper ferrules a half an inch wider than the flue sheet and the thickness through the water side is another half inch, then when you expand it, you do not crack it. In actual experience, we found with the wide copper ferrule you eliminate what we call "simmer" and at no time do you have a flue spurt, as you will with the narrow shim. When cold air hits it, with the narrow ferrule, it cracks and you have flue leakage, and with the wide copper ferrule, when cold air hits it, it contracts, shortens it and pulls it up to the joint made by the prosser expander and eliminates the simmer and helps your flue in several ways. By actual experience, I have found we decrease the failures and increase the mileage on our flues in many cases (without exaggerating), over sixty percent.

T. W. Lowe, Canadian Pacific: Our practice, on the Canadian Pacific, is that our coppers are cut the width of the tube sheets, and they are pushed into the sheets one-sixteenth short of the fire side of the sheet, allowing them to project one-sixteenth into the water. The purpose of allowing them to stand into the water, is that when we are driving in our flues, it prevents the copper from moving toward the fire. Therefore, when our tube is set, we are positive that the ferrule, which is hidden from view after setting, is not going to be exposed to the action of the gases and create a deterioration around your tube sheet hole. I notice, in the paper, that there is no reference made to the angularity of the rollers, as set in the body of the tube expander. This is one of the most important features of application, and furthermore, it is more important still when it comes to maintenance. This is true for the very reason that if you use a roller expander, set with a different angle, for maintenance, after you have applied your flues you are certainly going to shorten the life of your tube because the metal has been rolled from a certain position in the sheet and you oppose that and crumble the metal, and it is certainly very disastrous to the tube. In putting in flues, it is an advantage to roll them first. That is shop practice, but when we come to the roundhouse, one of our men, who had been studying application and maintenance, to overcome the difficulties that we were encountering, in the roundhouse, conceived the idea that he could get up a tool that would perform dual duties. These duties were that he could stretch the tube in the same operation as the beading of the tube, and he developed a tool which we have in use, today, in our roundhouses, performing these two functions.

H. J. Wandberg: Our practice on the Chicago, Milwaukee and St. Paul is that we clean holes, ream them or fill them as the case may be, and get them as near one size as possible, and apply our copper shims. The flue is cleaned off, the scale is taken off and it is made a driving fit and is driven in through the sheet, the proper length, and then turned over. We do not use any expander except the expander that we finish the flue with. We do not roll our flues in the firebox end. The roller is used for the front end, but not for the firebox end. We test all of our flues first. The reason for doing this is not to depend on the beading tool to make a flue tight. The expander must do that work. We have found, through experience, that lots of tubes are only half expanded, and then the beading tool made them tight, but it failed to do a good job. We have found that testing the boiler, before allowing the flue to be beaded, works out very well. We use a copper ferrule one-sixteenth of an inch wider than the thickness of our sheets. We have two thicknesses, ½ inch and ⅝ inch.

J. A. Anderson, Industrial Works: Fifty years ago I went into a boiler shop. That was before we, in that place, had ever seen a roller expander. The tubes were iron. The ordinary prosser expander has not been improved upon to any great extent, however, in that time. At that time, we had no trouble with flues, when we used the prosser expander. They were expanded with the prosser and beader. However, coming up to recent time, we made an experiment, recently, in the shop, by testing out the boiler before the flues were beaded. We had a special order from a railroad, who required that the flues should be welded at both ends. This was a boiler of sixty-five inches in diameter, and so they were applied, as the rule states here, by having the hole through, the edges sheared, and the copper ferrule a quarter of an inch wider and one-sixteenth of an inch back from the face of the sheet, and fastened by the expander. Then the flues were rolled and cleared about forty-five degrees, and then sand blasted and tested for beading, and we found scarcely and trouble with any flue, and, if there was, it was then beaded and welded and was a very satisfactory job.

President Lewis: Some men are using copper ferrules one-quarter of an inch wider than the sheet. Others are using copper ferrules one-eighth of an inch wider than the sheet, and one member said he was using copper ferrules almost flush, or a little narrower than the sheet. Now, which is better? We are using copper ferrules one-eighth of an inch wider than the flue sheet. If a quarter of an inch is better and gives better service under conditions where the flue service is so very severe and requires attention almost every trip, or once every month under some conditions, I would like to know about it.

D. A. Lucas: To get the best results, the flue has to be long enough so that it covers your sheet. If your flue is going to give you better service in bad water conditions, it surely is going to give you better service in good water conditions. Now, when our road extended up into the West, and went through the worst water conditions—alkali water—and you would take an engine out of the shops with a set of flues set in copper ⅝ inch wide, at that time, and send the engine up to that division, and from three to five months the engine had to come back to the shops on account of flues. The first set of flues put in with wide copper were made from a copper pipe. We could not buy wide copper at that time. We got a set of copper ferrules and put them in an inch wide, or a little better. There was not a lot of care taken in the cutting of their length. A set of flues were applied to that same engine, wide enough so that the expansion was made right in the copper, and that same engine went back to that division and stayed sixteen months without having the flues removed, or having any flue trouble. Now, if those conditions are better served with a wider flue, that ought to be enough to induce a man to use them, and to believe that there is something in the wide shim.

Charles P. Patrick: How do you swage your flues?

D. A. Lucas: They are swaged with gradual taper and all swaged when you use wide shim, and swaged far enough

back so that you have a gradual taper and then they stick tight—not an abrupt taper.

Charles P. Patrick: I want to know what kind of swage you use, the gradual?

D. A. Lucas: Gradual, yes, and turned two or three times to get it perfectly round, and, when you swage with that, it thickens the flue and maintains the full thickness of the flue, or a little better, all over. Our method was to prepare the flue sheets, clean out holes and in all cases see that there was a bevel on both sides of the flue sheet, to do away with the sharp edges. Then the flue sheet was applied, and rolled in lightly. Then the flue was driven in and then prossered with the sectional expander that had a taper on the outside that set the bead back to 30 degrees, ready for calking. It did away with that practice of turning the bead over. When you turn the bead over—lots of shops have a turning hammer—you hit the flue right in the neck, where you want the full thickness and, if you put in some flues and cut them, you will find your flue is reduced in the heel of the bead, where you want your strength.

Now, with this method that we use—and I have put in many flues and cut them, and then went into detail to find out the results—when the flue is expanded with this sectional expander, having a flare on the back to throw out flue (no hammer to throw it over), and the proper bead put on, you cut that flue open and you will find out that your bead is thicker right in the heel than in any other part of the flue.

President Lewis: I would like to know those who are using copper one-eighth of an inch wider than the sheet, and those who are using copper one-quarter of an inch wider than the sheet.

Mr. Glenway (Northern Pacific): Our copper extends three-sixteenths of an inch wider than the sheet.

W. H. Laughridge: I would state that we use one-eighth of an inch and get good results.

H. J. Wandberg: I may say that we use one-sixteenth of an inch, and have good results.

Mr. Howe (Burlington): We use one-half inch.

Mr. Mitchell: We use our coppers one-quarter of an inch wider than the sheet and our results are very satisfactory.

P. J. Conrath, National Tube Company: I would like to say that I have used the copper material wider than the flue sheet—one-half inch wider—for twenty-five years. The advantages claimed for the long copper come under two heads. First, you make an easier joint with the prosser on your expander in the shouldering of your flue sheet. Secondly, the closer the copper, if long enough around the back of the flue, the more prevention against sediment getting in between copper and the flue. That is the idea and that is the claim for the longer copper. Now, when your copper is made only one-eighth of an inch longer, it does no good. You simply create a mud pocket because it stands out of your copper only one-sixteenth of an inch and you catch it with the prosser of your expander and almost break the expander to set your flue. That is about all there is to these coppers being longer. Another thing about the setting of flues. I have set quite a few in my time. I have heard no one say where they begin to set flues and where end. That should be taken into consideration. Try to work the flues from the right to the left and you will find you have all the holes oblong. As soon as there is strain, and it gets hot, away goes the hole. Start a set of flues in a diamond in the center of the sheet. That is my method, but it might be a circle. Then work the flues all around so as to allow a place for your expansion in this pocket in the center. The sheet is flexible and will go wherever the expansion will drive it, which is not the case otherwise. If you work the flues all around into this pocket in the center, you keep this undue strain away from the flange.

A. N. Lucas: I believe a sixteenth wider than the sheet is the most satisfactory. We are using a light shim and it is all right. I believe you can use a shim, an expander in putting it in, and have it so that they will set the copper at least a thirty-second inch inside of the sheet, so that when the flue expands it will come out flush with the sheet, and there is no copper between the bead and sheet.

A. F. Stiglmeier: I would like to tell you that the majority of the railroads from whom I have information, are using ferrules ⅛ inch wider than the sheet. We have taken the majority on all points on which we secured information, and placed it into the report.

A. N. Lucas: I believe the flue subject is one of the best subjects that we have. The roundhouse should know how to take care of flues after being put in, in the proper manner, taking care of them in the right way.

P. J. Conrath: If the copper is only one-eighth of an inch wider than the sheet you turn them over, always. Otherwise you are going to have one long and one short, on account of rolling and the different ways of putting them in. The sheet of copper only one-eighth of an inch longer should be turned over against the sheet.

The Life of the Superheater Tube and the Number of Safe Ends That Should Be Applied Before Cutting Down for a Smaller Engine

IT is the consensus of opinion of your committee: That great care should be exercised so that as nearly as possible there will be only one weld on each flue. This can be accomplished by applying a five-inch safe end for weld No. 1; 6½-inch for No. 2; 7½-inch for No. 3; and 8½-inch for No. 4, thus cutting off each old weld before applying the new safe end. Welds on the front or large part of the flue are not desirable on account of applying units, etc.

We find it is the practice of some roads to weld on the large end after applying the limit of welds on the small end, while the practice of other roads is to apply only one weld before cutting down for a smaller boiler. This would seem to us a very expensive proposition if the locomotives are equipped with superheater tubes of the same length or nearly so.

We are also of the opinion that a safer weld can be obtained on a superheater tube than on a 2½ and 2¼ tube owing to the difference in the thickness of the metals.

This report was prepared by a committee consisting of J. P. Malley, chairman; H. Howard and Daniel S. Rice.

DISCUSSION

H. Howard, Illinois Central: We will all admit that we cannot determine the life of a flue, definitely, on account of the water. If we have good water, they will last a number of years, and, if there is bad water, the life is very short.

Andrew S. Green, Big Four: I do not believe you can put a 5-inch end on there, and the next time put a 6½-inch end, or your tube will be too short for the boiler. If you cut the ragged end off, it will be too short. You have to have more than 1 inch, there. I do not think that will work at all. If you put a 5-inch end on there, as you say, and next weld with 6½, you must have more metal or the tube grows too short for the boiler. You will have to have 7 inches on there.

A. N. Lucas: I believe that the best method of safe-ending is, after the tubes have made their first run, without any beads, to apply a 5 or 5½-inch piece at the first safe-end, and when it has made the mileage and comes in, you cut the tube again for welding. That is an extra operation that you can do in the fraction of a minute, and then put the next safe-ends on the opposite end and you still have left, when you cut the tube, enough to go into the boiler—about 4 inches of the first safe-end, and that will give you a good safe-end into the front end and avoid any cracking of the tube. The